Sunday, March 8, 2026 11:15 PM

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Example (4):

1000	COPY	Start	1000	
1000	FIRST	STL	RETADR	141033
1003	CLOOP	JSUB	RDREC	482039
1006		LDA	LENGTH	001036
1009		COMP	ZERO	281030
100C		JEQ	ENDFIL	301018
100F		JSUB	WRREC	
1012		J	CLOOP	
1015		LDA	EOF	
1018		STA	BUFFER	
101B	ENDFIL	LDA	THREE	00102D
101E		STA	LENGTH	001036
1021		JSUB	WRREC	
1024		LDL	RETADR	
1027		RSUB	4C0000	
102A	EOF	BYTE	C'EOF'	ASCII → 454F46
102D	THREE	WORD	3	000003
1030	ZERO	WORD	0	000000
1033	RETADR	RESW	1x3	
1036	LENGTH	RESW	1x3	NO OBJECT CODE
1039	BUFFER	RESB	4096x1	
2039	RDREC	LDX	ZERO	04 - - - - -
203C		LDA	ZERO	
203F	RLOOP	TD	INPUT	
2042		JEQ	RLOOP	
2045		RD	INPUT	
2048		COMP	ZERO	
204B		JEQ	EXIT	
204E		STCH	BUFFER	
2051		TIX	MAXLEN	
2054		JLT	RLOOP	
2057	EXIT	STX	LENGTH	
205A		RSUB	4C0000	
205D	INPUT	BYTE	X'F1'	→ F1
205E	MAXLEN	WORD	4096	→ 001000
2061	WRREC	LDX	ZERO	
2064	WLOOP	TD	OUTPUT	
2067		JEQ	WLOOP	
206A		LDCH	BUFFER,X	→ 509039
206D		WD	OUTPUT	
2070		TIX	LENGTH	
2073		JLT	WLOOP	
2076	OUTPUT	RSUB	4C0000	
2079		BYTE	X'05'	05
207A		END		

Symbol Table

COPY	1000
FIRST	1000
CLOOP	1003
ENDFIL	101B
EOF	102A
ZERO	1030
RETADR	1033
THREE	102D
LENGTH	1036
BUFFER	1039
RDREC	2039
RLOOP	203F
EXIT	2057
INPUT	205D
MAXLEN	205E
WRREC	2061
WLOOP	2064
OUTPUT	2076

H.COPYXX.001000.00107A

T.001000.1E.- - - - - - -

T.00101E.15.001036.- - - - - - - - -

T.002039.1E.- - - - - - - - -

T.002057.1C.- - - - - - -

T.- - - - - -

E.001000

H COPY 001000 00107A

T 001000 1E 141033 482039 001036 281030 301015 482061 3C1003 00102A 0C1039 00102D

T 00101E 15 0C1036 482061 081033 4C0000 454F46 000003 000000

T 002039 1E 041030 001030 E0205D 30203F D8205D 281030 302057 549039 2C205E 38203F

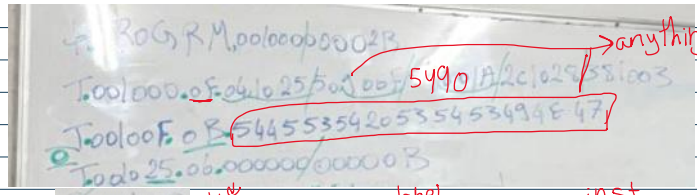
T 002057 1C 101036 4C0000 F1 001000 041030 E02079 302064 509039 DC2079 2C1036

T 002073 07 382064 4C0000 05

T 002057 1C 101056 460000 +1 001000 041030 E02014 302064 509039 DC2014 2C1036

T 002073 07 382064 460000 05

E 001000



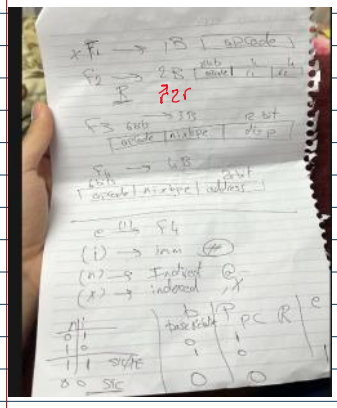
anything > 7 → ,x → subtract 8

Decimal values

Hexa	label	inst	object code
1000	PROGRM	START	1000
1000		LDX	A_1
1003	A_5	LDCH	A_2,X
1006		STCH	A_3,X
1009		TIX	A_4
100C		JLT	A_5
100F	A_2	BYTE	x"[illegible]"
101A	A_3	RESB	11
1025	A_1	word	0
1028	A_4	word	11
102B		End	PROGRAM

Symbol Table

A_1	1025
A_2	100F
A_3	101A
A_4	1028
A_5	1003



1 → B 8
2 → 2B 16
3 → 3B 24
4 → 4B 32

n → @ → indirect
i → # → immediate
x → ,X → indexed
b → baserelative → 0 ≤ disp ≤ 4095
p → pc relative → -2048 ≤ disp ≤ 2047
e → + → extended

e=1 → f4
n,i = 1 → SIC/XE
n,i = 0 → SIC

(arks) Dis-assemble the following SICXE HTE record to complete the corresponding highlighted parts of the program. Also
is of the symbol

is also provided

0000	EXAM	START	0000
0000		LDX	#0
0003		+LDB	#SUM
007		BASE	SUM
		LDS	#GOAL
000A	L1	LDA	SUM
0017		STA	AX,X
1A		+TIXR	JARD
1E			
002F		STA	GOAL
0031	L2	RSUB	
0034	N	WORD	(X'00')
0037	GOAL	RESW	1
003A	AX	RESW	300
0BF2	SUM	WORD	0
0BF5	END	XXEXAM	

Symbol	Location
L1	000A
L2	0031
GOAL	0037
AX	003A

Sum 0BF2

Byte X'00', X'0C', X'0

mixope

0110 1101 0010 / 02D

0110 1100 → 6C

HEXEXAM 0000030000B5

T000000 0A 05 0000 4B 3300 3F 2FD 2D 2D

T000003 0D 03 4000 3B 2002 4F 0000 4F 0000 20 1D

T000017 1E 07 A020 8810 0310 0B F2 A0 4037200C03200C190001

T00002B 0B 07 F200 3F 2FD 4C0000

T000034 0C 0000 01 0000DC

T000BF2 0000000

M00000105 → -1

M00001305 → -1

M00000005 → -1

E000000

020

FD1 = TA - 0051

100A